

Physical AI and the magnet supply chain

The physics and the mathematics behind the supplier decisions

A lesson for an investor who reads papers · Compiled 22 September 2026

Companion to the explainer doc *Physical AI and the magnet supply chain — investor explainer*. Educational research only; not investment advice.

What this document does. It derives, rather than asserts, the six results that decide who wins the race to put permanent magnets into humanoid robots: why $\text{Nd}_2\text{Fe}_{14}\text{B}$ has no cheap rival, why motor heat forces dysprosium and terbium into the magnet, why coercivity is a defect property that grain-boundary diffusion can exploit, why the joint architecture of a robot is also its rare-earth bill, and why separating adjacent lanthanides costs so many stages that a 98% Chinese share in heavy separation is chemistry as much as policy. Each chapter ends with the consequence for a supplier decision.

How the numbers were produced. Every curve in the figures is computed by `scripts/lesson/magnetics.py`, which has a unit test per function (`scripts/tests/test_lesson_magnetics.py`). Every literature constant is cited to a paper or page opened on 22 September 2026; the bibliography lists only sources verified through the Crossref DOI registry, a publisher page, or the open-access text. Where a number is a modelling assumption rather than a measurement it is labelled as such in an orange *Label* box.

Notation. SI units. $\mathbf{B}$ is flux density (T), $\mathbf{H}$ the magnetic field (A/m), $\mathbf{M}$ the magnetisation (A/m) and $\mathbf{J} = \mu_0 \mathbf{M}$ the polarisation (T); $\mu_0 = 4\pi \times 10^{-7}$ H/m. Fields quoted in tesla are $\mu_0 H$. Coercivity: H_{cJ} is where J crosses zero (intrinsic coercivity), H_{cB} where B does. 1 kOe = 79.58 kA/m; 1 MGOe = 7.958 kJ/m³ [1].

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1 From tokens to torque: why physical AI is a metallurgy problem

Digital AI is limited by compute, memory and electricity. Physical AI takes the same transformer models, feeds them cameras and language, and has them emit joint torques instead of tokens. The brain stays in silicon; the body is 20 to 50 electric motors, and every one of them is built around a sintered Nd–Fe–B permanent magnet [2]. A humanoid carries 2 to 4.5 kg of such magnet, the lower figure for high-gear-ratio joints and the higher for quasi-direct-drive joints, which works out to 0.57 to 1.3 kg of neodymium–praseodymium and 35 to 79 g of dysprosium per robot under the assumption of H- and SH-grade magnets with 50% dysprosium thrifting [2].

Those four numbers (magnet mass, gear ratio, magnet grade, dysprosium thrifting) are not marketing choices. Each is fixed by a physical relation that this document derives:

Decision	Physics that fixes it	Chapter
Which magnet material	Energy product bound $(BH)_{\max} \leq B_r^2/4\mu_0$ and the anisotropy field $H_A = 2K_1/\mu_0 M_s$	2, 3
Which grade (N, H, SH, UH)	Load line versus the temperature-shifted knee of the $J(H)$ curve	4
How much Dy or Tb	Coercivity is set by nucleation at grain surfaces: $H_c = \alpha H_A - N_{\text{eff}} M_s$; diffusion depth $\propto \sqrt{Dt}$	5
Joint architecture	Air-gap shear stress: $\tau = 2\pi r_g^2 l \sigma$; magnet mass $\propto \tau/N$	6
Who can refine the heavy elements	Separation factor β near 1 and the Fenske stage count $N \propto 1/\ln \beta$	7

Investor consequence. The supply chain that matters is not “rare earths”. It is a specific crystal ($\text{Nd}_2\text{Fe}_{14}\text{B}$), a specific defect-engineering process (grain-boundary diffusion of Dy or Tb), and a specific chemistry (hundreds of solvent-extraction stages). A company’s position in the race is its position on those three, not on a mine.

2 Magnetostatics of a permanent magnet

2.1 Fields, magnetisation and the demagnetising field

Inside matter the three field quantities are related by

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0\mathbf{H} + \mathbf{J}. \quad (1)$$

Two of Maxwell's equations govern a magnet with no free current: $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{H} = 0$. The second means $\mathbf{H}$ is conservative, so around any closed loop $\oint \mathbf{H} \cdot d\mathbf{l} = 0$. The first, applied to a uniformly magnetised body, forces $\mathbf{H}$ inside the body to point *against* $\mathbf{M}$: the magnet is the source of a field that opposes its own magnetisation. For an ellipsoid this demagnetising field is uniform,

$$\mathbf{H}_d = -N\mathbf{M}, \quad (2)$$

with N the demagnetising factor: $1/3$ for a sphere, 1 for a thin plate magnetised through its thickness, and 0 for a long rod magnetised along its axis [3], [4]. A thin magnet magnetised through its thickness, which is exactly the shape of a motor magnet, sits in the worst possible self-field. That geometric fact is the root of every thermal problem in Chapter 4.

2.2 The magnetic circuit and the load line

Consider a magnet of length l_m and cross-section A_m driving flux across an air gap of length l_g and area A_g through an ideal iron yoke (infinite permeability, no leakage). Ampère's law around the circuit, with no current, gives

$$H_m l_m + H_g l_g = 0, \quad (3)$$

and flux continuity gives $B_m A_m = B_g A_g$ with $B_g = \mu_0 H_g$ in the gap. Eliminating the gap quantities:

$$B_m = -\mu_0 \frac{l_m A_g}{l_g A_m} H_m \equiv -\mu_0 P_c H_m. \quad (4)$$

Equation Equation 4 is the *load line*: a straight line through the origin of the second quadrant of the B - H plane whose slope, $-\mu_0 P_c$, is fixed entirely by geometry. P_c is the permeance coefficient. A long thin magnet in a short gap has a high P_c and operates near B_r ; a thin plate in a wide gap has a low P_c and operates far down its curve, in a large reverse field. The operating point is the intersection of the load line with the magnet's own $B(H)$ curve [5], [6].

The “infinite permeability” yoke is the one idealisation an investor should not forget. Real electrical steel saturates: the flux density in a stator tooth cannot usefully exceed the saturation polarisation of iron, about 2 T [4], and a motor is normally designed with its teeth close to that limit. Once the steel is saturated, extra remanence from a better magnet grade buys nothing; the flux has nowhere to go. In a steel-limited design a grade upgrade is bought for coercivity, i.e. for temperature margin (Chapter 4), not for torque. Whether a given motor is magnet-limited or steel-limited is a question a supplier can answer in one sentence and a spec sheet cannot.

2.3 Why the energy product is the figure of merit

The magnetostatic energy stored in the gap is $B_g^2 V_g / 2\mu_0$, with $V_g = A_g l_g$. Work with twice that quantity (the factor $1/2$ would appear on both sides and cancel). Using Equation 3 and flux continuity,

$$B_g^2 \frac{V_g}{\mu_0} = (B_g A_g)(H_g l_g) = (B_m A_m)(-H_m l_m) = (-B_m H_m) V_m. \quad (5)$$

So the magnet volume needed to hold a given field in a given gap is

$$V_m = \frac{B_g^2 V_g}{\mu_0 (-B_m H_m)}. \quad (6)$$

The magnet is smallest when the product $-BH$ at the operating point is largest. That product, maximised over the second quadrant, is the energy product $(BH)_{\max}$ [1]. Everything a motor designer buys from a magnet maker is compressed into that one number and the temperature at which it survives.

2.4 Anatomy of the loop

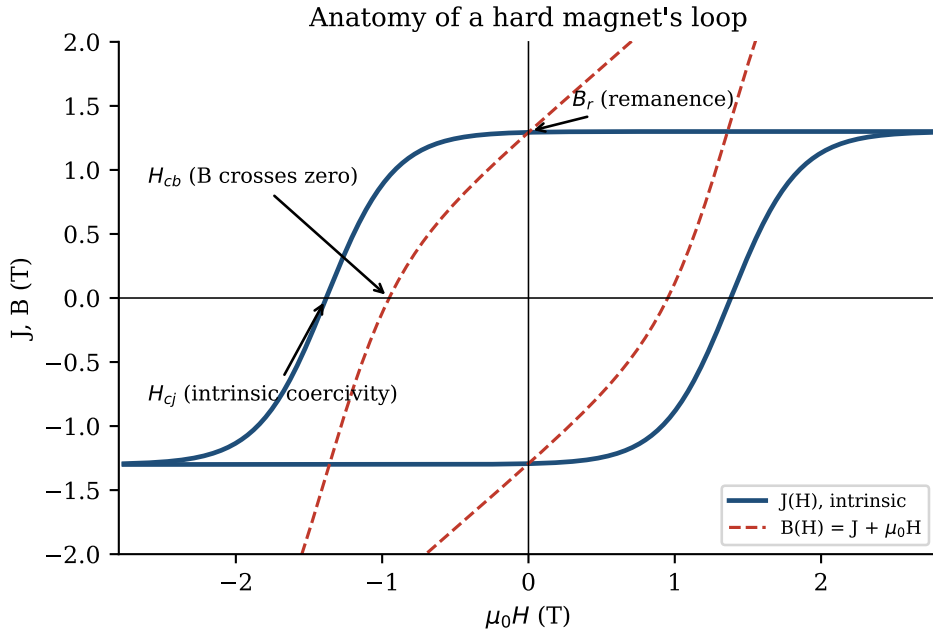


Figure 1: The intrinsic loop $J(H)$ (solid) and the normal loop $B(H) = J + \mu_0 H$ (dashed) of a hard magnet. B_r is the remanence, H_{cJ} the field that reverses the polarisation, H_{cB} the smaller field at which B itself crosses zero. Computed from a smooth model loop; the shape, not the numbers, is the lesson.

The intrinsic curve $J(H)$ is the material; the normal curve $B(H)$ is what the circuit sees. For a well-made magnet J stays flat at B_r deep into the second quadrant and then collapses at the *knee*, near H_{cJ} . Along the flat part, with a small recoil permeability μ_r close to 1,

$$B(H) = B_r + \mu_0 \mu_r H, \quad H \leq 0, \quad (7)$$

so the energy product along the line is $-BH = -(B_r + \mu_0 \mu_r H)H$, maximised at $H = -B_r/(2\mu_0 \mu_r)$:

$$(BH)_{\max} = \frac{B_r^2}{4\mu_0 \mu_r} \leq \frac{B_r^2}{4\mu_0}. \quad (8)$$

The bound is reached only if the knee lies beyond the optimum point, i.e. if $\mu_0 H_{cJ} > B_r/2$ roughly. A magnet with a high remanence but a low coercivity never reaches its own bound: this is why alnico, whose remanence is comparable to Nd-Fe-B's but whose coercivity is small, never comes close to its own bound [4].

Worked number. For $\text{Nd}_2\text{Fe}_{14}\text{B}$ the room-temperature saturation polarisation is $J_s = 1.60 \text{ T}$ [7]. Equation 8 with $B_r = J_s$ and $\mu_r = 1$ gives $(1.60)^2 / (4 \times 4\pi \times 10^{-7}) = 509 \text{ kJ/m}^3$, exactly the theoretical maximum listed by Hirose et al. [7]. Commercial sintered grades reach 200 to 440 kJ/m^3 [8] because real remanence is below J_s (grain misalignment, non-magnetic grain-boundary phase, porosity) and because the knee truncates the curve at high temperature.

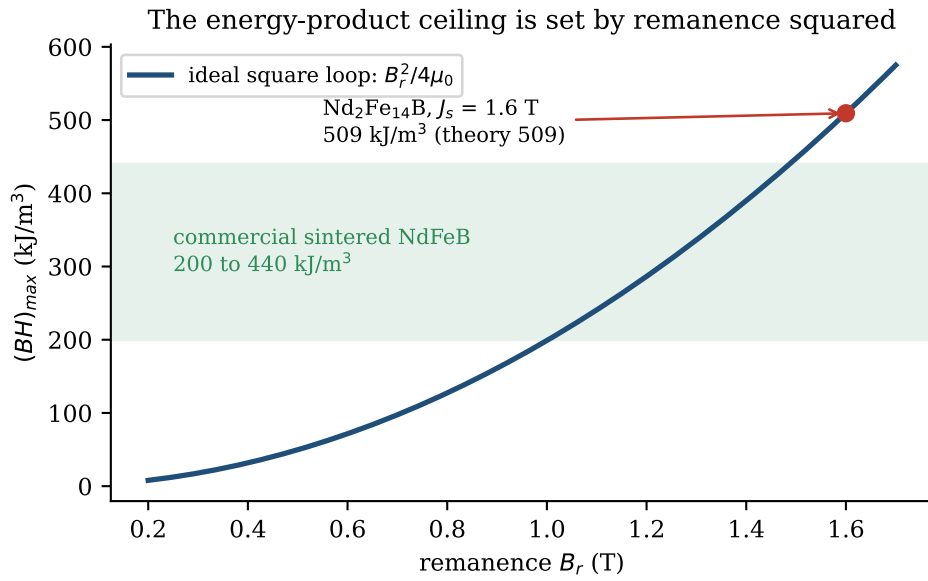


Figure 2: The energy-product ceiling $B_r^2/4\mu_0$ against remanence. The $\text{Nd}_2\text{Fe}_{14}\text{B}$ point reproduces the 509 kJ/m^3 theoretical value; the band is the commercial sintered range [8].

Investor consequence. The magnet is a volume-per-torque device. A 10% loss of remanence (from heat, from cerium substitution, from a cheaper grade) is a 19% loss of energy product by Equation 8, and by Equation 6 a 23% larger magnet to do the same job. Small changes in B_r are large changes in the bill of materials.

3 Why Nd₂Fe₁₄B has no cheap rival

3.1 Where the remanence comes from

The compound discovered in 1984 [9], [10] is tetragonal Nd₂Fe₁₄B: 14 iron atoms carry most of the moment, and the two neodymium atoms add to it because the light rare-earth 4f moment couples parallel to the iron moment. The result is a saturation polarisation of 1.60 T, a saturation magnetisation $M_s = 1.28$ MA/m, and a Curie temperature of 586 K [7], [11]; the single-crystal magnetisation and anisotropy of the whole R₂Fe₁₄B family were mapped in the two years after the discovery [12], [13]. The heavy rare earths (Gd onward) couple *antiparallel* to the iron, which is why substituting them lowers the remanence, a point Chapter 4 returns to.

3.2 Where the coercivity comes from: magnetocrystalline anisotropy

A ferromagnet's energy depends on the direction of $\mathbf{M}$ relative to the crystal axes. For a uniaxial crystal with easy axis c and angle θ between $\mathbf{M}$ and c ,

$$E_a = K_1 \sin^2 \theta + K_2 \sin^4 \theta + \dots, \quad (9)$$

and for Nd₂Fe₁₄B at room temperature $K_1 = 4.3$ MJ/m³ [7]. The physical origin is the strongly aspherical 4f charge cloud of the Nd³⁺ ion sitting in the crystal's electric field; spin-orbit coupling locks the Fe–Nd spin system to the orientation that cloud prefers [14], [15]. Density-functional calculations overestimate K_1 (about 8 MJ/m³ against the measured 4.3) because thermal disorder of the 4f moments already matters at 300 K [11], which is a first hint that anisotropy, and hence coercivity, is the temperature-fragile property.

The field needed to rotate $\mathbf{M}$ away from the easy axis against K_1 is the *anisotropy field*. Write the energy per unit volume with a field of magnitude H applied *against* $\mathbf{M}$, so that the Zeeman term is $+\mu_0 M_s H \cos \theta$: $E = K_1 \sin^2 \theta + \mu_0 M_s H \cos \theta$. The aligned state $\theta = 0$ stops being a minimum when $\partial^2 E / \partial \theta^2 |_{\theta=0} = 2K_1 - \mu_0 M_s H = 0$, which gives

$$H_A = \frac{2K_1}{\mu_0 M_s}. \quad (10)$$

Worked number. With $K_1 = 4.3$ MJ/m³ and $M_s = 1.28$ MA/m, Equation 10 gives $H_A = 5.35$ MA/m, i.e. $\mu_0 H_A = 6.7$ T. Hirosawa et al. list 5.33 MA/m [7]. The 1984 pulsed-field measurement gave a larger 12 MA/m for Nd₂Fe₁₄B and found the maximum anisotropy field of the whole R₂Fe₁₄B series in Tb₂Fe₁₄B at 28 MA/m [16]; the absolute values depend on the extrapolation method, but the ratio Tb : Nd of about 2.3 by the same method is the number that matters for Chapter 5.

3.3 The Stoner–Wohlfarth limit

For a single-domain particle whose magnetisation rotates coherently, the energy is $E = K_u V \sin^2(\varphi - \theta) - \mu_0 M_s V H \cos \varphi$ [17], [18]. With the field antiparallel to the easy axis, the particle switches exactly at $H = H_A$; for a random assembly of such particles the coercivity is $0.48 H_A$ [18]. This is the ceiling: no magnet can have a coercivity above its anisotropy field, and a perfect aligned magnet would switch at H_A itself. Real sintered Nd–Fe–B switches at a fifth of that (Chapter 5).

3.4 The competition, in the same units

Material	Why it loses to Nd–Fe–B	Where it still wins	Source
Hard ferrite	Energy product about 1/18 of sintered Nd–Fe–B by volume:	Cost; positive β (coercivity rises with temperature)	[8], [19]

	the ferrite motor for the same torque is several times heavier		
Sm-Co (2:17)	Lower J_s so lower $(BH)_{\max}$; cobalt and samarium cost; samarium export-licensed since April 2025	Curie temperature 720 °C; small α and β ; radiation and corrosion resistance	[8], [19], [20]
Alnico	Coercivity of a few tens of kA/m: the knee truncates Equation 8	Temperature stability	[19]
Ce-substituted Nd-Fe-B	For $x = 0.38$ Ce on the Nd site, H_A at 400 K falls from 5.5 to 4.7 T and T_C from 586 to 543 K [11]	Cerium is the cheap by-product of every Nd mine	[11]
Rare-earth-free motors	Induction and reluctance machines exist [21] but carry a torque-density penalty that a 2 t car can hide and a 30 kg robot cannot	Cost, supply security	[21]

The wider landscape of hard, soft and energy-conversion magnetic materials is reviewed in [22]. Nanostructured exchange-spring magnets (hard phase plus soft high-moment phase) were predicted to exceed 1 MJ/m³ [23] and remain a laboratory result thirty years later; the gap between ferrite and Nd-Fe-B, the “plugging the gap” problem [24], is still open.

Investor consequence. There is no substitution story at the top of the performance range: every alternative in the table above loses on remanence, on coercivity, or on both, and the energy-product bound Equation 8 is squared in remanence. The substitution stories that are real are *downward*: cerium for cost, ferrite where torque density does not matter. A robot joint is the application where torque density matters most, so it is the last place substitution arrives.

4 Heat: why the grade ladder exists and why it runs on dysprosium

4.1 Reversible temperature coefficients

Between room temperature and the working temperature both remanence and coercivity fall approximately linearly:

$$B_r(T) = B_r(T_0)[1 + \alpha(T - T_0)], \quad H_{cJ}(T) = H_{cJ}(T_0)[1 + \beta(T - T_0)]. \quad (11)$$

For sintered Nd-Fe-B, $\alpha \approx -0.12 \text{ } \%/^{\circ}\text{C}$ and $\beta \approx -0.55$ to $-0.65 \text{ } \%/^{\circ}\text{C}$; for Sm-Co, $\alpha \approx -0.04$ and $\beta \approx -0.2$ to -0.4 ; ferrite has $\alpha \approx -0.20$ but a *positive* $\beta \approx +0.27 \text{ } \%/^{\circ}\text{C}$ [19]. Hirosawa et al. quote about $-0.5 \text{ } \%/K$ for the coercivity of Nd-Dy-Fe-B [7]. The asymmetry is the whole story: at $120 \text{ } ^{\circ}\text{C}$ an Nd-Fe-B magnet has lost 12% of its remanence but 60% of its coercivity. Remanence loss costs torque; coercivity loss costs the magnet its life.

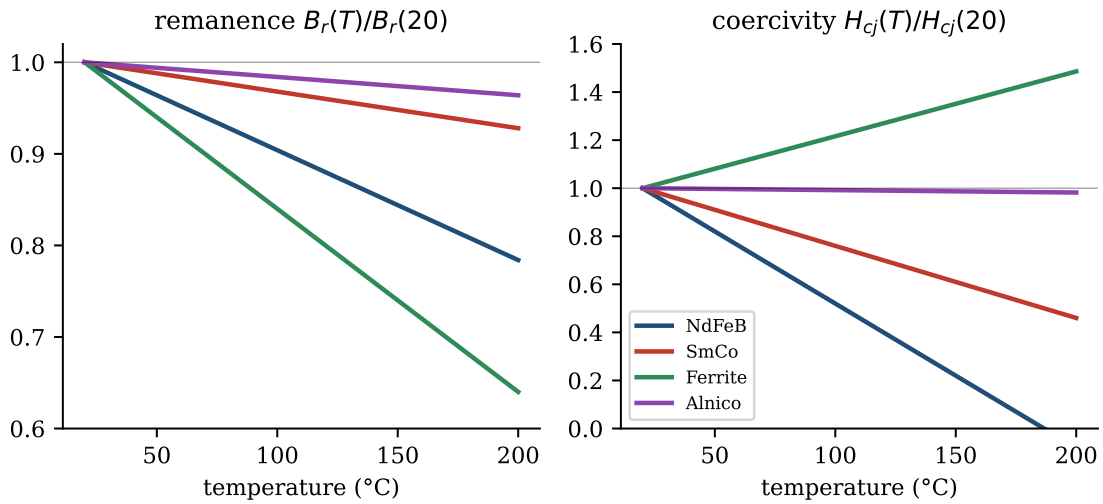


Figure 3: Normalised remanence and coercivity against temperature from Equation 11 with the coefficients in [19]. Nd-Fe-B's coercivity collapses fastest; ferrite's rises.

4.2 The knee crossing

The operating point sits where the load line Equation 4 crosses $B(H)$. As the magnet heats, B_r falls a little and the knee marches toward the origin. When the knee reaches the load line, the polarisation at the operating point starts to collapse and the loss is *irreversible*: cooling the magnet does not restore it, because the reversed grains stay reversed. The design rule is therefore not “stay below the Curie temperature” (586 K) but “keep the operating field a safe margin inside the knee at the hottest point of the duty cycle”.

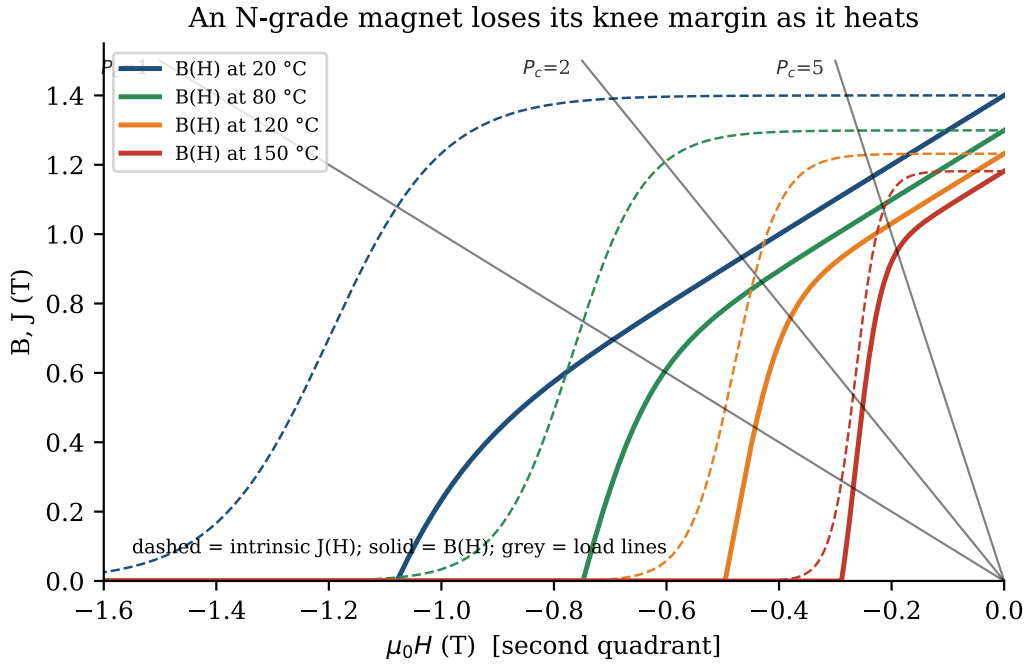


Figure 4: Second-quadrant curves of an N-type magnet (B_r 1.40 T, H_{cJ} 12 kOe at 20 °C, $\alpha = -0.12$, $\beta = -0.60$ %/°C) at four temperatures, with load lines for $P_c = 1, 2$ and 5. By 120 °C the knee has crossed the $P_c = 1$ and 2 lines. The knee shape is a smooth model; the coefficients are from [19] and [25].

Label. The “highest safe temperature” curves below use a toy criterion: the operating field on the load line, times a 20% margin, must stay below $H_{cJ}(T)$. Manufacturers rate grades more conservatively (they use the actual knee field, which lies below H_{cJ} , and specific test geometries), so the absolute temperatures here run higher than nominal ratings. The *ordering* and the *slope* against P_c are the physics; the absolute values are the model.

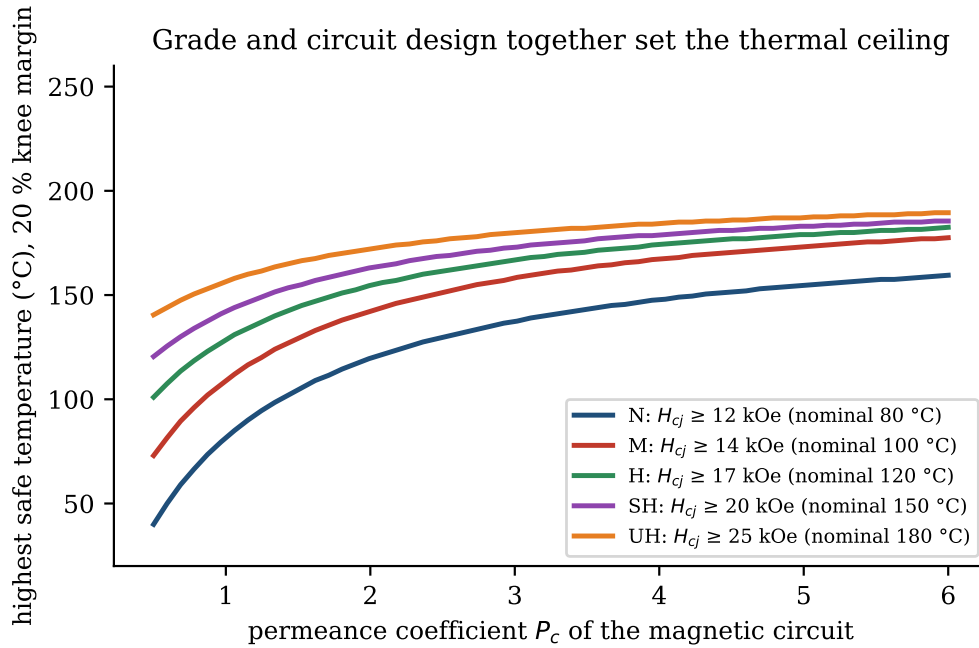


Figure 5: Highest safe temperature against permeance coefficient for five grades, from `max_safe_temperature()` with the minimum coercivities of the class table [25]. Two levers move the ceiling: the grade (vertical spacing) and the circuit design (slope).

Worked number. At $P_c = 2$ the toy criterion gives 120 °C for N (12 kOe), 155 °C for H (17 kOe), 163 °C for SH (20 kOe) and 172 °C for UH (25 kOe). The nominal class ratings are 80, 120, 150 and 180 °C [25]. Each class step (N to M, M to H, H to SH, and so on) adds 3 to 5 kOe of coercivity and 20 to 30 °C of nominal rating [25], and each step costs more dysprosium or terbium.

4.3 Why the fix is dysprosium

Coercivity is proportional to the anisotropy field, for reasons Chapter 5 derives (Equation 12). The heavy rare earths raise H_A : by the 1984 single-method comparison, $\text{Tb}_2\text{Fe}_{14}\text{B}$ has about 2.3 times the anisotropy field of $\text{Nd}_2\text{Fe}_{14}\text{B}$ [16], so substituting Dy or Tb for part of the Nd pushes the knee outward at every temperature; the proportionality of coercivity to anisotropy field in Dy-substituted sintered magnets was established directly in 1987 [26], and the intrinsic properties of the $(\text{Nd,Dy})_2\text{Fe}_{14}\text{B}$ series are still being refined in high-field work [27]. Per gram, terbium is the more effective of the two: about 2.5 g of dysprosium does the coercivity work of 1 g of terbium [2], and terbium is the scarcer and dearer element.

The cost is remanence. The heavy rare-earth 4f moment couples antiparallel to the iron sublattice, so each Dy atom subtracts moment: in $(\text{Nd}_{2-x}\text{Dy}_x)\text{Fe}_{14}\text{B}$ the formula-unit moment falls from $25.50 \mu_B$ at $x = 0$ to 23.48 at $x = 0.25$ and 21.03 at $x = 0.5$, while coercivity roughly doubles [28]. Dy occupies the 4f rare-earth site specifically [28]. By Equation 8 the remanence penalty is squared in the energy product. This trade, coercivity up and energy product down, is the whole reason the industry moved from alloying Dy through the bulk to placing it only where it is needed.

Investor consequence. A magnet grade is a dysprosium content, and the grade is chosen by the motor's hottest duty point on the load line. Two suppliers can move the bill: the magnet maker, by getting the same coercivity with less Dy (Chapter 5), and the motor maker, by raising P_c or improving cooling so a lower grade survives (Chapter 6). Ask both which lever they own.

5 Coercivity is a defect property: Brown's paradox and grain-boundary diffusion

5.1 Brown's paradox

The Stoner–Wohlfarth ceiling says a perfect aligned crystal reverses only at H_A : 5.3 MA/m for Nd₂Fe₁₄B. A commercial N-grade sintered magnet reverses at 12 kOe, i.e. 0.95 MA/m, less than a fifth of H_A . This shortfall, general to all hard magnets, was pointed out by Brown in 1945 [29]: reversal does not happen by coherent rotation of a whole grain but by *nucleation* of a reversed domain at some defect where the local anisotropy is reduced or the local field is enhanced, followed by domain-wall propagation. Once one grain reverses, it seeds its neighbours unless something isolates them.

Kronmüller's analysis of nucleation in inhomogeneous ferromagnets [30], applied to Nd–Fe–B by Kronmüller, Durst and Sagawa [31], condenses the microstructure into two parameters:

$$H_c = \alpha H_A - N_{\text{eff}} M_s. \quad (12)$$

The form follows from two superpositions. First, nucleation happens where it is easiest: in a surface layer whose anisotropy constant is reduced to $K_{1'} < K_1$ (disorder, oxidation, a few nanometres of off-stoichiometry), the Stoner–Wohlfarth switching field of that layer is $2K_{1'}/\mu_0 M_s = \alpha_K H_A$ with $\alpha_K = K_{1'}/K_1 < 1$; misalignment of the grain axis to the field lowers it by a further factor α_ψ , and $\alpha = \alpha_K \alpha_\psi$. Second, the field that actually acts at a grain edge is the applied field plus the local stray field of the grain's own magnetisation, $-N_{\text{eff}} M_s$; reversal begins when their sum reaches the reduced switching field, so the applied field needed is αH_A *minus* the stray-field help [30], [31]. Both α and N_{eff} are properties of the grain *surface* and its geometry, not the grain interior.

Worked number. Take $H_{cJ} = 0.95$ MA/m, $H_A = 5.35$ MA/m, $M_s = 1.28$ MA/m and, illustratively, $N_{\text{eff}} = 1$: Equation 12 gives $\alpha = (0.95 + 1.28)/5.35 = 0.42$. The fitted values in the literature vary with grain size and processing; the point is that α sits far below 1, and that the term $N_{\text{eff}} M_s$ costs a full MA/m. At these values the reduced surface anisotropy accounts for about 70% of the shortfall from H_A (3.1 MA/m) and the stray-field term for about 30% (1.3 MA/m).

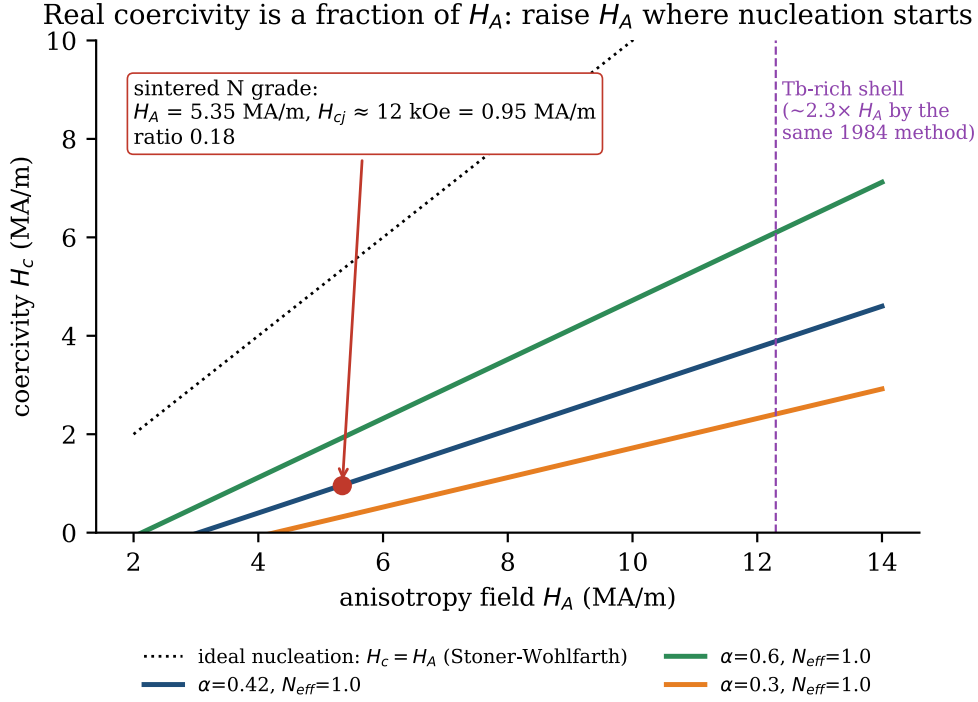


Figure 6: Coercivity against anisotropy field in the Kronmüller form Equation 12 for three values of α , against the Stoner-Wohlfarth line. The N-grade point sits at a ratio of 0.18. The dashed line marks the anisotropy field of a Tb-rich shell, about 2.3 times Nd's by the same measurement [16].

Equation Equation 12 is the whole coercivity strategy of the last twenty years in one line [32]:

1. *Raise H_A at the surface*, where nucleation starts, with Dy or Tb. Because $H_c \propto \alpha H_A$, a shell with 2.3 times the anisotropy field multiplies the coercivity of the grain even if the core is untouched.
2. *Isolate the grains* with a non-ferromagnetic grain-boundary phase so a reversed grain cannot seed its neighbour. Atom-probe work showed the Nd-rich boundary phase in a standard sintered magnet is thin and itself ferromagnetic, which is why isolation is imperfect [33].
3. *Shrink the grains* so the stray-field term $N_{eff}M_s$ at edges is smaller; Nakamura et al. showed that very small sintered magnets, and by extension the surface treatment that became grain-boundary diffusion, gain coercivity this way [34].

5.2 Grain-boundary diffusion

Instead of alloying Dy through the bulk (which pays the remanence penalty of Chapter 4 everywhere), grain-boundary diffusion (GBD) coats a finished magnet with a Dy or Tb compound (fluoride, oxide, or a low-melting Dy-Ni-Al alloy [35]), then anneals it below the sintering temperature. The heavy element travels along the liquid Nd-rich grain boundaries far faster than through the grains and substitutes into the outer layer of each $\text{Nd}_2\text{Fe}_{14}\text{B}$ grain, forming a core-shell structure with a $(\text{Nd,Dy})_2\text{Fe}_{14}\text{B}$ or $(\text{Nd,Tb})_2\text{Fe}_{14}\text{B}$ shell of high H_A around a Dy-free core of full remanence [32], [36], [37]. One vendor description quotes more than a 50% reduction of the coercivity loss at 150 °C for such magnets [37].

The transport is Fick's second law, $\partial C / \partial t = D \partial^2 C / \partial x^2$, and for a constant surface source in a semi-infinite body the solution is [38]

$$C(x, t) = C_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right). \quad (13)$$

The penetration depth scales as $\sqrt{Dt}$: doubling the depth costs four times the furnace time. This is why GBD works on thin parts. The uniform core-shell layer in a Tb-diffused magnet lies about 300 to 1000 μm below

the surface [36], and extending useful treatment to thicker plates required a dedicated low-melting diffusion source [35].

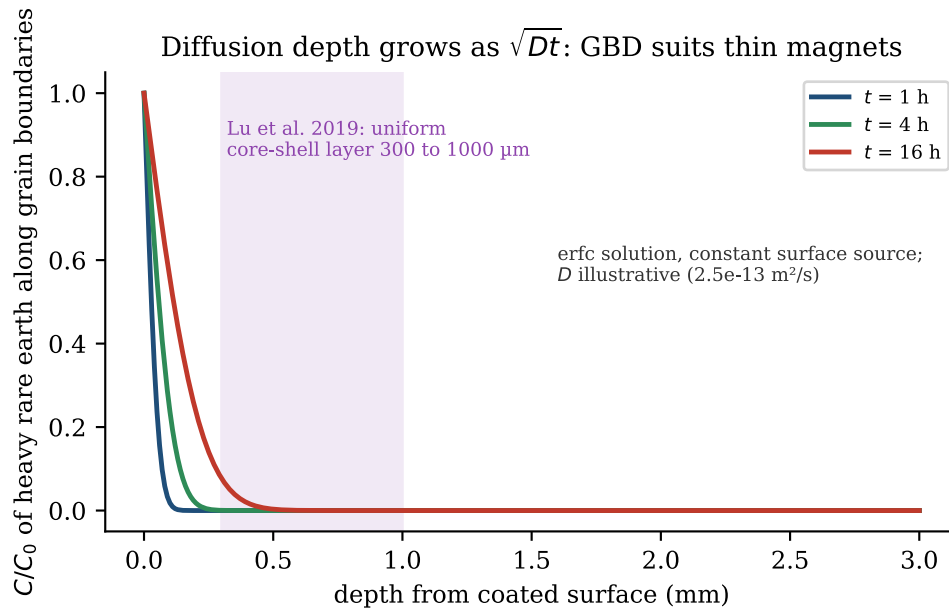


Figure 7: Heavy rare-earth concentration along grain boundaries against depth from Equation 13 for three anneal times with an illustrative effective diffusivity. The band marks the depth range of the uniform core-shell layer reported by Lu et al. [36].

Label. The diffusivity in the figure is illustrative, chosen to place the shell at the reported depth; it is not a measured grain-boundary diffusion coefficient. The $\sqrt{Dt}$ scaling is exact; the constant is not.

Investor consequence. GBD is the reason heavy rare-earth demand per magnet can halve without changing the magnet's rating, which is the "50% dysprosium thrifting" behind every per-robot estimate [2]. It works best on thin magnets, and motor magnets are thin arc segments: the process and the application fit. Whether a Western magnet plant runs GBD in-house, under licence, or not at all decides whether it makes a competitive SH/UH magnet or an expensive one. Ask for the coercivity per gram of Dy, not the coercivity.

6 The motor: why the joint architecture is the rare-earth bill

6.1 Torque from shear stress in the air gap

A conductor of length l carrying current I in a field B feels the Lorentz force $F = BIl$. Spread the stator conductors around the air gap as a linear current density K_s (ampere-turns per metre of circumference) facing a gap flux density B_g from the rotor magnets: the tangential force per unit gap area is a *shear stress* $\sigma = B_g K_s$, and the torque on a rotor of gap radius r_g and active length l is the stress times the gap area times the lever arm,

$$\tau = \sigma(2\pi r_g l) r_g = 2\pi r_g^2 l \sigma. \quad (14)$$

This is the sizing equation of every permanent-magnet machine [5], and the form Wensing et al. use for actuator scaling [39]. Two things follow immediately. Torque per unit rotor volume is 2σ , independent of size. And B_g , which is proportional to the magnet's remanence at the operating point, enters σ linearly: a 10% loss of B_r is a 10% loss of torque at the same current, or, at the same torque, $1/0.9$, i.e. 11% more current and $(1/0.9)^2 - 1 = 23\%$ more copper loss.

6.2 The thermal limit sets the shear stress

Copper loss per unit gap area is proportional to K_s^2 (resistance times current squared per metre of periphery). At a fixed cooling capacity, K_s is therefore capped, and with it the shear stress: $\sigma_{\max} = B_g K_{s,\max}$. For a family of motors of the same construction and cooling, σ is constant and Equation 14 says $\tau \propto r_g^2 l$ [39]. Torque per unit mass then scales as r_g in the dimensional analysis; the catalogue data Wensing et al. fitted give $r_g^{0.8}$ for torque density, $r_g^{1.6}$ for torque per rotor inertia and $r_g^3 l$ for the torque-production efficiency $\tau^2 / I^2 R$ [39]. Larger gap radius means more torque per kilogram and far more torque per unit inertia, at the cost of a bigger, heavier joint.

One loss mechanism sits inside the magnet itself. Sintered Nd-Fe-B is a metal, so the slot harmonics and switching ripple of the stator induce eddy currents in the rotor magnets and heat them, precisely the component whose coercivity is the most temperature-fragile (Chapter 4). Designers answer by segmenting each pole into several insulated pieces, which cuts the eddy loss but multiplies the machining, coating and assembly cost per pole [5]. The same segmentation makes each piece thinner, which is what grain-boundary diffusion wants (Chapter 5): the loss-driven and the coercivity-driven reasons for small magnets point the same way.

6.3 Magnet mass per joint and the gear ratio

A joint needs torque τ_j . Behind a reducer of ratio N the motor supplies $\tau_m = \tau_j / N$. At constant shear stress the rotor volume, and so the magnet volume on it, is proportional to τ_m :

$$m_{\text{magnet}} \propto \frac{\tau_j}{N}. \quad (15)$$

A harmonic or cycloidal joint needs an order of magnitude less motor, and magnet, than a quasi-direct-drive (QDD) joint for the same joint torque; the ratios used below, $N \approx 50$ –100 for the geared joint and $N \approx 6$ –10 for QDD, are typical values chosen for illustration, not measured on a particular robot. The published per-robot figures, 2 kg for high-ratio designs against 4.5 kg for QDD [2], differ by less than Equation 15 alone predicts because QDD designs also use larger gap radii (Equation 14: torque $\propto r_g^2$) and thinner magnets, but the direction is the same and it is not small.

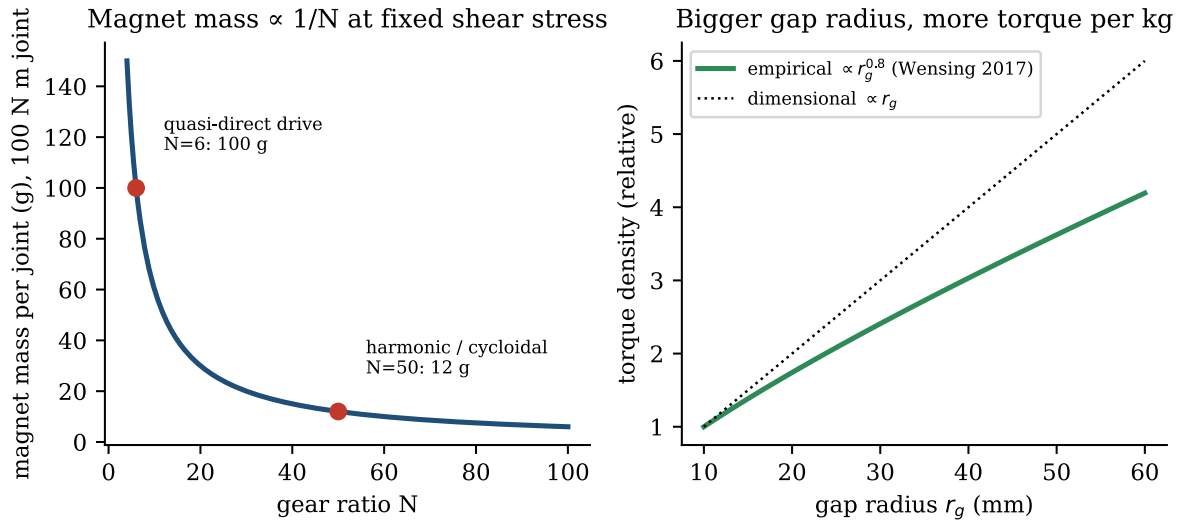


Figure 8: Left: magnet mass per joint against gear ratio from Equation 15 for a 100 N m joint (constant of proportionality illustrative). Right: torque density against gap radius, dimensional (r_g^1) and empirical ($r_g^{0.8}$) [39].

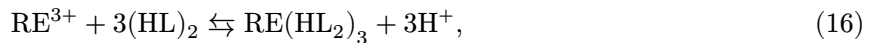
Why would anyone choose QDD and pay ten times the magnet? Because the reflected inertia of the motor at the joint scales as N^2 , and a high-ratio joint is stiff, slow to back-drive and destructive on impact. Wensing et al. define an *impact mitigation factor* from the ratio of the reflected actuator inertia to the locked-joint inertia [39]; a low ratio lets the leg absorb a foot strike and lets the controller feel contact through motor current (proprioception) instead of through a fragile force sensor. Walking robots want QDD at the hips and knees; manipulation wants it at the wrists. The magnet bill follows the biomechanics.

Investor consequence. The robot maker's choice of reducer is its choice of magnet mass, and the two are public: a QDD humanoid needs roughly twice the magnet of a geared one [2]. Watch the reducer suppliers (harmonic and cycloidal drives are a second, Japan-centred chokepoint) as closely as the magnet suppliers, and read every torque-density claim for the gap radius it assumes.

7 Separating the heavy elements: why 98% is chemistry

7.1 The lanthanide contraction and the separation factor

The fifteen lanthanides [40] differ by one 4f electron each, and 4f electrons are buried inside the ion; the ions are all trivalent and their radii shrink steadily and only slightly along the row (the lanthanide contraction). Chemically they are near-identical, which is why they were once called “rare earths” and were separated by hundreds of fractional crystallisations. Modern separation is liquid–liquid extraction [41]: an aqueous chloride or nitrate solution is mixed with an organic phase containing an acidic organophosphorus extractant (D2EHPA/P204, or 2-ethylhexyl phosphonic acid mono-2-ethylhexyl ester, EHEHPA/P507), the metal partitions into the organic phase by a reaction of the form



and the two phases are separated. The distribution ratio $D = [\text{RE}]_{\text{org}}/[\text{RE}]_{\text{aq}}$ rises steeply with pH and with extractant concentration, and rises with atomic number. The selectivity between two elements A and B is the *separation factor*

$$\beta_{A/B} = \frac{D_A}{D_B}. \quad (17)$$

For adjacent lanthanides β is close to 1. Ismail et al. tabulate the values for both industrial extractants in chloride medium [42]:

Pair	Pr/Nd	Nd/Sm	Sm/Eu	Eu/Gd	Gd/Tb	Tb/Dy	Dy/Ho	Ho/Er	Er/Tm	Tm/Yb	Yb/Lu
P507 (EHEHPA)	1.17	2.00	1.96	1.46	2.35	1.62	2.58	1.25	1.33	1.12	1.13
P204 (D2EHPA)	1.06	4.86	2.23	1.69	1.60	1.42	1.24	1.70	1.50	1.30	1.03

Two pairs matter for magnets: Pr/Nd, which is so hard (1.17 with P507, 1.06 with P204) that industry sells the pair together as “didymium” NdPr, and Tb/Dy at 1.62, which is separable but sits in a row of heavy elements each of which must be split from the next.

7.2 Counting the stages

One equilibrium contact enriches the organic phase in the heavier element by a factor β . To go from a 50:50 feed to 99.9% purity at both ends requires a cascade of contacts run counter-currently. The stage count follows from one line of algebra that does not care whether the two phases are vapour and liquid or organic and aqueous. At an ideal equilibrium stage the ratio of the two elements in the extract is β times their ratio in the feed to that stage, by Equation 17. In a counter-current cascade at total reflux every stage’s product is the next stage’s feed, so after n stages $(x_A/x_B)_n = \beta^n (x_A/x_B)_0$. Solving for the n that takes the ratio from its value at the bottom to its value at the top is the Fenske relation, first written for distillation [43] with the relative volatility in the place of β :

$$N_{\min} = \frac{\ln \left[\left(\frac{x_{\text{top}}}{1-x_{\text{top}}} \right) \left(\frac{1-x_{\text{bottom}}}{x_{\text{bottom}}} \right) \right]}{\ln \beta}. \quad (18)$$

Worked number. For 99.9% purity at both ends the numerator is $\ln(999^2) = 13.8$. With P507: Pr/Nd, $\beta = 1.17$, needs at least 88 stages; Tb/Dy, $\beta = 1.62$, 29 stages; Ho/Er, $\beta = 1.25$, 62 stages; Dy/Ho, $\beta = 2.58$, 15 stages. Real cascades run at finite reflux, add scrubbing and stripping sections, and must split a feed of a dozen elements into individual products, so plant stage counts are multiples of these lower bounds.

Label. The stage counts assume ideal equilibrium stages, a separation factor that is constant along the cascade, and total reflux. Real β drifts with concentration and acidity, and a working cascade runs at finite reflux with scrubbing and stripping sections, so these are lower bounds, not plant designs.

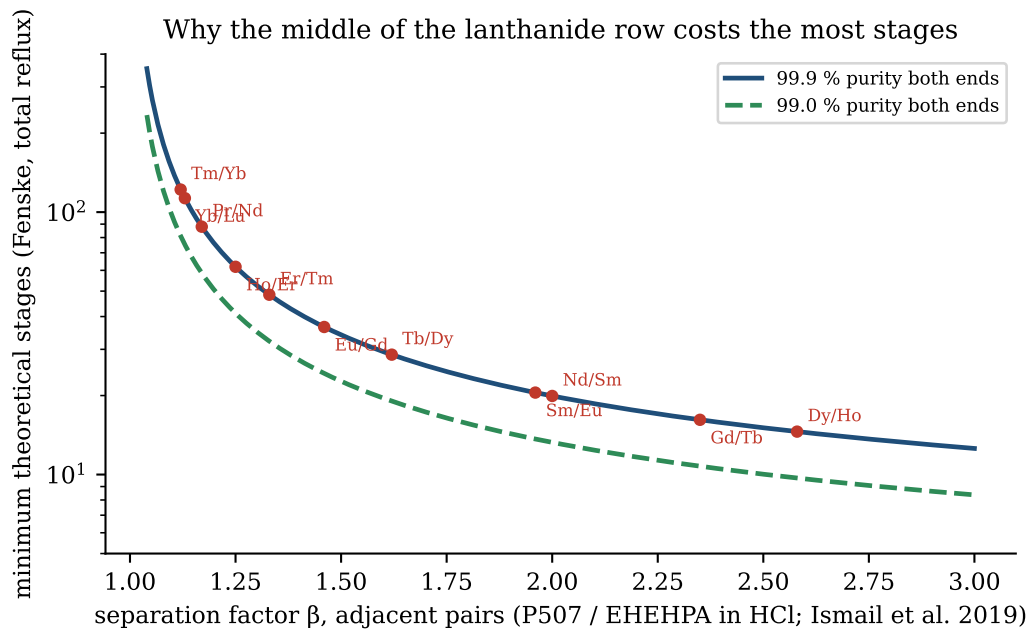


Figure 9: Minimum theoretical stages from Equation 18 against separation factor, with the adjacent-pair values for P507 [42]. The count diverges as $\beta \rightarrow 1$; the light Pr/Nd pair and the middle-heavy pairs are the expensive ones.

7.3 Why the heavy end is where the moat is

Three things compound at the heavy end. The feed is small: the world sold 2.7 million kg of dysprosium and 0.8 million kg of terbium in 2024 against 87 million kg of NdPr [2], so a heavy-rare-earth plant is a small plant with a long cascade, and its economics depend on running every stage full. The chemistry is slower: heavy elements bind the extractant more strongly, which makes stripping harder and pushes producers to blended extractants (P507 with Cyanex 272) whose selectivities are proprietary know-how [42], [44]. And the feedstock is geographically concentrated: more than 90% of heavy rare-earth supply originates from ion-adsorption clays [45], and China refines about 98% of heavy rare earths [45] against 91% of all refining and 94% of sintered magnets [46]. Lynas produced the first commercial dysprosium oxide outside China only in May 2025, and its first terbium in June 2025 [47], [48]; Energy Fuels targets 48 t of Dy oxide and 14 t of Tb oxide a year from late 2026 [49].

Investor consequence. A Western heavy-separation plant is a stage-count problem before it is a mining problem: a Tb/Dy split alone is a 30-stage-plus cascade sized for tens of tonnes a year. Judge a project by its stage count, its extractant chemistry and its feed contract, not by its resource statement. The light end is the easier half of the business for three reasons that the table makes explicit: the Pr/Nd split is simply not attempted (the pair is sold as didymium), the Nd/Sm boundary has $\beta = 2.00$ with P507, and the volumes are thirty times larger so a cascade runs full. The Ce/Pr boundary ($\beta = 1.09$) is handled in practice by chemistry outside this table, which is not covered here. That is why the West already refines NdPr and why China's April 2025 list left it alone [20].

8 Synthesis: the arithmetic and the decisions

8.1 The dysprosium arithmetic

Per robot, 35 to 79 g of dysprosium [2]. Per million robots a year, 35 to 79 t. That is 1.3 to 2.9% of the 2,700 t sold worldwide in 2024, and less than the 1,470 t of Dy-equivalent metal the 2025 electric-vehicle fleet consumed [2]. But it is of the same order as the entire funded Western capacity: 48 t of Dy oxide from Energy Fuels from late 2026 [49], rising to about 120 t in 2027–28 and 288 t in 2028–29 if both expansion phases are built [50]. The scarcity is regional, not global.

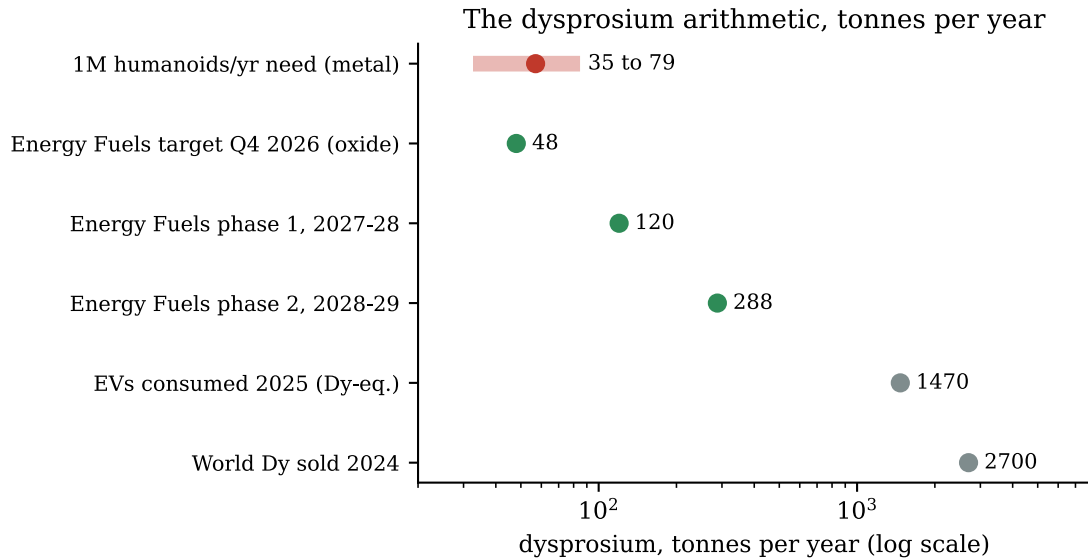


Figure 10: Dysprosium, tonnes a year, on a log axis: robot demand per million units (metal, [2]), the Western oxide-capacity steps [49], [50], EV consumption and world sales [2]. Oxide and metal tonnes are not identical; the comparison is indicative.

8.2 Each derivation, one decision

Result	What it says	What to ask a supplier
Equation 8, Equation 6	Magnet volume $\propto 1/B_r^2$	What remanence at the operating temperature, not at 20 °C?
Equation 4, Figure 4	The knee crossing, not the Curie point, kills the magnet	What P_c does the motor run at, and what is the hottest duty point?
Equation 11	Coercivity falls five times faster than remanence	Which grade, and what is the coercivity margin at that duty point?
Equation 12	Coercivity is set at grain surfaces	Coercivity per gram of Dy or Tb; is GBD in-house?
Equation 13	GBD depth $\propto \sqrt{Dt}$	What magnet thickness does the line handle?
Equation 14, Equation 15	Magnet mass $\propto \tau_j/N$; torque $\propto r_g^2$	Which joints are QDD, and who makes the reducers?
Equation 18	Stages $\propto 1/\ln \beta$	Stage count, extractant, and the heavy-feed contract

8.3 What would change this document

Three developments would move the conclusions, and each is observable. A dysprosium-free joint motor at humanoid torque density, whether by a thermally superior motor design (raising P_c or cooling until an H grade suffices), by an exchange-spring or iron-nitride magnet leaving the laboratory [23], [24], or by a rare-earth-free machine accepting the weight penalty [21]. A Western grain-boundary-diffusion line at scale, since it halves the heavy-element bill at a stroke. And a heavy-separation cascade outside China running at full utilisation, which is a stage-count and feed-contract question, not a geology question. Recycling of end-of-life magnets [51] is the only heavy-element supply that can arrive in months rather than years, and it feeds directly into the magnet-to-magnet processes that Western entrants are building.

9 Appendix: reproducibility and labels

Code. `scripts/lesson/magnetics.py` implements every relation used here: `bh_max_ideal`, `demag_curve`, `anisotropy_field`, `kronmuller_hc`, `operating_point`, `at_temperature`, `max_safe_temperature`, `fenske_stages`, `erfc_profile`, `torque_from_shear`, `magnet_mass_for_joint`. The tests in `scripts/tests/test_lesson_magnetics.py` pin the worked numbers (509 kJ/m³ 5.35 MA/m against the literature 5.33 within 1%; 88 stages). `python -m lesson.figures` regenerates every figure and prints the quoted values; `typst compile lesson.typ` rebuilds this PDF.

Verified versus modelled. Verified (cited): all material constants, temperature coefficients, grade classes, separation factors, per-robot and market tonnages, and the scaling exponents. Modelled (labelled in orange boxes): the knee shape of the $J(H)$ curve, the 20% knee margin and the resulting absolute safe temperatures, the effective diffusivity in Figure 7, and the constant of proportionality in Figure 8. Not verified this session and therefore not quoted: absolute anisotropy fields of Dy₂Fe₁₄B and Tb₂Fe₁₄B from modern single-crystal work (the 1984 same-method ratio is used instead [16]), and the Kronmüller fit parameters of specific commercial grades.

Disclaimer. Educational and research material. Not investment advice. The companion explainer doc carries the market, policy and company data with their own sources; this document carries the physics.

References

- [1] Wikipedia contributors, “Maximum energy product.” Accessed: Sept. 22, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Maximum_energy_product
- [2] Foundation for American Innovation, “The robot boom has a magnet problem.” Accessed: Sept. 22, 2026. [Online]. Available: <https://www.thefai.org/posts/the-robot-boom-has-a-magnet-problem>
- [3] Wikipedia contributors, “Demagnetizing field.” Accessed: Sept. 22, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Demagnetizing_field
- [4] B. D. Cullity and C. D. Graham, *Introduction to Magnetic Materials*, 2nd ed. Wiley, 2009.
- [5] J. R. Hendershot and T. J. E. Miller, *Design of Brushless Permanent-Magnet Machines*. Motor Design Books, 2010.
- [6] J. M. D. Coey, *Magnetism and Magnetic Materials*. Cambridge University Press, 2010.
- [7] S. Hirosawa, M. Nishino, and S. Miyashita, “Perspectives for high-performance permanent magnets: applications, coercivity, and new materials,” *Advances in Natural Sciences: Nanoscience and Nanotechnology*, vol. 8, p. 13002, 2017, doi: [10.1088/2043-6254/aa597c](https://doi.org/10.1088/2043-6254/aa597c).
- [8] Wikipedia contributors, “Neodymium magnet.” Accessed: Sept. 22, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Neodymium_magnet
- [9] M. Sagawa, S. Fujimura, N. Togawa, H. Yamamoto, and Y. Matsuura, “New material for permanent magnets on a base of Nd and Fe (invited),” *Journal of Applied Physics*, vol. 55, no. 6, pp. 2083–2087, 1984, doi: [10.1063/1.333572](https://doi.org/10.1063/1.333572).
- [10] J. J. Croat, J. F. Herbst, R. W. Lee, and F. E. Pinkerton, “Pr-Fe and Nd-Fe-based materials: A new class of high-performance permanent magnets (invited),” *Journal of Applied Physics*, vol. 55, no. 6, pp. 2078–2082, 1984, doi: [10.1063/1.333571](https://doi.org/10.1063/1.333571).
- [11] M. A. Susner *et al.*, “Growth and characterization of Ce-substituted Nd₂Fe₁₄B single crystals,” 2015. doi: [10.48550/arXiv.1508.07792](https://doi.org/10.48550/arXiv.1508.07792).
- [12] S. Hirosawa, Y. Matsuura, H. Yamamoto, S. Fujimura, M. Sagawa, and H. Yamauchi, “Magnetization and magnetic anisotropy of R₂Fe₁₄B measured on single crystals,” *Journal of Applied Physics*, vol. 59, pp. 873–879, 1986, doi: [10.1063/1.336611](https://doi.org/10.1063/1.336611).
- [13] R. Grössinger, X. K. Sun, R. Eibler, K. H. J. Buschow, and H. R. Kirchmayr, “Temperature dependence of anisotropy fields and initial susceptibilities in R₂Fe₁₄B compounds,” *Journal of Magnetism and Magnetic Materials*, vol. 58, pp. 55–60, 1986, doi: [10.1016/0304-8853\(86\)90122-8](https://doi.org/10.1016/0304-8853(86)90122-8).
- [14] D. Haskel *et al.*, “Atomic origin of magnetocrystalline anisotropy in Nd₂Fe₁₄B,” *Physical Review Letters*, vol. 95, 2005, Accessed: Sept. 22, 2026. [Online]. Available: <https://www3.aps.anl.gov/haskel/PS/ndfeb.pdf>
- [15] J. F. Herbst, “R₂Fe₁₄B materials: Intrinsic properties and technological aspects,” *Reviews of Modern Physics*, vol. 63, no. 4, pp. 819–898, 1991, doi: [10.1103/RevModPhys.63.819](https://doi.org/10.1103/RevModPhys.63.819).
- [16] H. Yamamoto, Y. Matsuura, S. Fujimura, and M. Sagawa, “Magnetocrystalline anisotropy of R₂Fe₁₄B tetragonal compounds,” *Applied Physics Letters*, vol. 45, pp. 1141–1143, 1984, doi: [10.1063/1.95015](https://doi.org/10.1063/1.95015).
- [17] E. C. Stoner and E. P. Wohlfarth, “A mechanism of magnetic hysteresis in heterogeneous alloys,” *Philosophical Transactions of the Royal Society of London. Series A*, vol. 240, pp. 599–642, 1948, doi: [10.1098/rsta.1948.0007](https://doi.org/10.1098/rsta.1948.0007).
- [18] Wikipedia contributors, “Stoner-Wohlfarth model.” Accessed: Sept. 22, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Stoner%E2%80%93Wohlfarth_model
- [19] Bunting Magnetics, “Temperature coefficients of magnet materials.” Accessed: Sept. 22, 2026. [Online]. Available: <https://www.magnetapplications.com/blog/temperature-coefficients-of-magnets>

- [20] China Briefing, “China’s rare earth export controls: impact on businesses and industries.” Accessed: Sept. 22, 2026. [Online]. Available: <https://www.china-briefing.com/news/chinas-rare-earth-export-controls-impacts-on-businesses/>
- [21] J. D. Widmer, R. Martin, and M. Kimiabeigi, “Electric vehicle traction motors without rare earth magnets,” *Sustainable Materials and Technologies*, vol. 3, pp. 7–13, 2015, doi: [10.1016/j.susmat.2015.02.001](https://doi.org/10.1016/j.susmat.2015.02.001).
- [22] O. Gutfleisch, M. A. Willard, E. Brück, C. H. Chen, S. G. Sankar, and J. P. Liu, “Magnetic materials and devices for the 21st century: Stronger, lighter, and more energy efficient,” *Advanced Materials*, vol. 23, pp. 821–842, 2011, doi: [10.1002/adma.201002180](https://doi.org/10.1002/adma.201002180).
- [23] R. Skomski and J. M. D. Coey, “Giant energy product in nanostructured two-phase magnets,” *Physical Review B*, vol. 48, pp. 15812–15816, 1993, doi: [10.1103/PhysRevB.48.15812](https://doi.org/10.1103/PhysRevB.48.15812).
- [24] J. M. D. Coey, “Permanent magnets: Plugging the gap,” *Scripta Materialia*, vol. 67, no. 6, pp. 524–529, 2012, doi: [10.1016/j.scriptamat.2012.04.036](https://doi.org/10.1016/j.scriptamat.2012.04.036).
- [25] Radial Magnets, “Neodymium magnet temperature classes: N, M, H, SH, UH, EH explained.” Accessed: Sept. 22, 2026. [Online]. Available: <https://radialmagnet.com/neodymium-magnet-temperature-classes/>
- [26] M. Sagawa *et al.*, “Dependence of coercivity on the anisotropy field in the Nd₂Fe₁₄B-type sintered magnets,” *Journal of Applied Physics*, vol. 61, no. 8, pp. 3559–3561, 1987, doi: [10.1063/1.338725](https://doi.org/10.1063/1.338725).
- [27] N. V. Kostyuchenko *et al.*, “High-field magnetization study of (Nd,Dy)₂Fe₁₄B: Intrinsic properties and promising compositions,” *Intermetallics*, vol. 124, p. 106840, 2020, doi: [10.1016/j.intermet.2020.106840](https://doi.org/10.1016/j.intermet.2020.106840).
- [28] S. K. Haider, M.-C. Kang, J. Hong, Y. S. Kang, C.-W. Yang, and D. Kim, “Determination of Dy substitution site in Nd_{2-x}Dy_xFe₁₄B by HAADF-STEM and illustration of magnetic anisotropy of g and f sites, before and after substitution,” *Scientific Reports*, vol. 11, p. 6347, 2021, doi: [10.1038/s41598-021-85713-5](https://doi.org/10.1038/s41598-021-85713-5).
- [29] W. F. Brown, “Virtues and weaknesses of the domain concept,” *Reviews of Modern Physics*, vol. 17, pp. 15–19, 1945, doi: [10.1103/RevModPhys.17.15](https://doi.org/10.1103/RevModPhys.17.15).
- [30] H. Kronmüller, “Theory of nucleation fields in inhomogeneous ferromagnets,” *physica status solidi (b)*, vol. 144, no. 1, pp. 385–396, 1987, doi: [10.1002/pssb.2221440134](https://doi.org/10.1002/pssb.2221440134).
- [31] H. Kronmüller, K.-D. Durst, and M. Sagawa, “Analysis of the magnetic hardening mechanism in RE-FeB permanent magnets,” *Journal of Magnetism and Magnetic Materials*, vol. 74, no. 3, pp. 291–302, 1988, doi: [10.1016/0304-8853\(88\)90202-8](https://doi.org/10.1016/0304-8853(88)90202-8).
- [32] K. Hono and H. Sepehri-Amin, “Strategy for high-coercivity Nd-Fe-B magnets,” *Scripta Materialia*, vol. 67, pp. 530–535, 2012, doi: [10.1016/j.scriptamat.2012.06.038](https://doi.org/10.1016/j.scriptamat.2012.06.038).
- [33] H. Sepehri-Amin, T. Ohkubo, T. Shima, and K. Hono, “Grain boundary and interface chemistry of an Nd-Fe-B-based sintered magnet,” *Acta Materialia*, vol. 60, no. 3, pp. 819–830, 2012, doi: [10.1016/j.actamat.2011.10.043](https://doi.org/10.1016/j.actamat.2011.10.043).
- [34] H. Nakamura, K. Hirota, M. Shimao, T. Minowa, and M. Honshima, “Magnetic properties of extremely small Nd-Fe-B sintered magnets,” *IEEE Transactions on Magnetics*, vol. 41, no. 10, pp. 3844–3846, 2005, doi: [10.1109/TMAG.2005.854874](https://doi.org/10.1109/TMAG.2005.854874).
- [35] N. Oono, M. Sagawa, R. Kasada, H. Matsui, and A. Kimura, “Production of thick high-performance sintered neodymium magnets by grain boundary diffusion treatment with dysprosium-nickel-aluminum alloy,” *Journal of Magnetism and Magnetic Materials*, vol. 323, pp. 297–300, 2011, doi: [10.1016/j.jmmm.2010.09.021](https://doi.org/10.1016/j.jmmm.2010.09.021).
- [36] Y. Lu *et al.*, “Nd-Fe-B magnets: The gradient change of microstructures and the diffusion principle after grain boundary diffusion process,” *Materials*, vol. 12, no. 23, p. 3881, 2019, doi: [10.3390/ma12233881](https://doi.org/10.3390/ma12233881).
- [37] Stanford Magnets, “The complete guide to high-temperature neodymium magnets.” Accessed: Sept. 22, 2026. [Online]. Available: <https://www.stanfordmagnets.com/complete-guide-to-high-temperature-neodymium-magnets.html>

- [38] J. Crank, *The Mathematics of Diffusion*, 2nd ed. Oxford University Press, 1975.
- [39] P. M. Wensing, A. Wang, S. Seok, D. Otten, J. Lang, and S. Kim, “Proprioceptive actuator design in the MIT Cheetah: Impact mitigation and high-bandwidth physical interaction for dynamic legged robots,” *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 509–522, 2017, doi: [10.1109/TRO.2016.2640183](https://doi.org/10.1109/TRO.2016.2640183).
- [40] Wikipedia contributors, “Rare-earth element.” Accessed: Sept. 22, 2026. [Online]. Available: https://en.wikipedia.org/wiki/Rare-earth_element
- [41] F. Xie, T. A. Zhang, D. Dreisinger, and F. Doyle, “A critical review on solvent extraction of rare earths from aqueous solutions,” *Minerals Engineering*, vol. 56, pp. 10–28, 2014, doi: [10.1016/j.mineng.2013.10.021](https://doi.org/10.1016/j.mineng.2013.10.021).
- [42] N. A. Ismail, M. A. Abd Aziz, M. Y. Mohd Yunus, and A. Hisyam, “Selection of extractant in rare earth solvent extraction system: A review,” *International Journal of Recent Technology and Engineering*, vol. 8, no. 1, pp. 728–743, 2019, Accessed: Sept. 22, 2026. [Online]. Available: <https://www.ijrte.org/wp-content/uploads/papers/v8i1/F2887037619.pdf>
- [43] M. R. Fenske, “Fractionation of straight-run Pennsylvania gasoline,” *Industrial & Engineering Chemistry*, vol. 24, no. 5, pp. 482–485, 1932, doi: [10.1021/ie50269a003](https://doi.org/10.1021/ie50269a003).
- [44] V. C. Arellano Ruiz, R. Kuchi, P. K. Parhi, J.-Y. Lee, and R. K. Jyothi, “Environmentally friendly comprehensive hydrometallurgical method development for neodymium recovery from mixed rare earth aqueous solutions using organo-phosphorus derivatives,” *Scientific Reports*, vol. 10, p. 16911, 2020, doi: [10.1038/s41598-020-74041-9](https://doi.org/10.1038/s41598-020-74041-9).
- [45] Rare Earth Exchanges, “Heavy rare earths: the 98% refining problem no one can ignore.” Accessed: Sept. 22, 2026. [Online]. Available: <https://rareearthexchanges.com/news/heavy-rare-earths-the-98-refining-problem-no-one-can-ignore/>
- [46] International Energy Agency, “Rare earth elements: executive summary.” Accessed: Sept. 22, 2026. [Online]. Available: <https://www.iea.org/reports/rare-earth-elements/executive-summary>
- [47] Mining.com.au, “Lynas achieves first terbium production in Malaysia.” Accessed: Sept. 22, 2026. [Online]. Available: <https://mining.com.au/lynas-achieves-first-terbium-production-in-malaysia/>
- [48] Center for Strategic and International Studies, “Rare earth export restrictions one year later.” Accessed: Sept. 22, 2026. [Online]. Available: <https://www.csis.org/analysis/rare-earth-export-restrictions-one-year-later>
- [49] Energy Fuels Inc., “Energy Fuels’ US-produced heavy rare earth oxide successfully qualified for use in permanent magnets.” Accessed: Sept. 22, 2026. [Online]. Available: <https://investors.energyfuels.com/2025-12-19-Energy-Fuels-US-Produced-Heavy-Rare-Earth-Oxide-Successfully-Qualified-for-Use-in-Permanent-Magnets>
- [50] Discovery Alert, “Energy Fuels White Mesa Mill expansion: what you need to know.” Accessed: Sept. 22, 2026. [Online]. Available: <https://discoveryalert.com/energy-fuels-white-mesa-heavy-rare-earth-expansion-2026/>
- [51] K. Binnemans *et al.*, “Recycling of rare earths: a critical review,” *Journal of Cleaner Production*, vol. 51, pp. 1–22, 2013, doi: [10.1016/j.jclepro.2012.12.037](https://doi.org/10.1016/j.jclepro.2012.12.037).